



Mark Scheme (Results)

June 2025

Pearson BTEC Nationals

In Information Technology (31760H)

Unit 1: Information Technology Systems

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

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Question Number	Answer	Mark
1ai	Any two from: Lossy (1) Lossless (1) Accept any other phrasing/wording	2

Question Number	Answer	Mark
1bi	Award up to two marks for each of two linked explanations, from: Faster / quicker speed (1) which makes uploading/ downloading/ sending/receiving data quicker (1) Improved security (1) (provides a fixed connection) which makes hacking more difficult / to gain access the user has to have physical access to a network port (1) More reliable / stable (1) because they are not affected by environmental / external factors / less susceptible to interference / signal strength does not deteriorate over distance (1) Additional guidance: Responses can be reversed, if they gave a disadvantage of wireless connection DO NOT ACCEPT: 'quicker' or 'faster' on its own. Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
1bii	Award up to two marks for each of two linked explanations, from: • Less portability/mobility for users (1) because the laptop has to be plugged in to the router (1) • Takes time / time consuming (1) because cables will need to be installed (1) • Trip hazards (1) because of cables not being tidied away properly (1) • Cost implications (1) because wired networks require extra costs for cables / hire a specialist to set up (1) • Limited amount of devices can connect (1) because there are a few number of ports available (1) Additional guidance: Responses can be reversed. Learners might give a disadvantage of wireless connection Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
1c	Award up to two marks for each of two linked explanations, from: • Nino could lose signal / signal may drop out (1) which would disconnect him from the (company network) / he would not be able to access his work (1) • Bandwidth may be affected (1) which means the upload/download time may increase/decrease (1) • Cost implications (1) because if he goes over his data allowance/plan and will need to purchase more (1) • Mobile data limit has been reached/ran out (1) connection is lost (1) Accept any other relevant phrasing/wording	4

Question Number	Indicative content
1d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the characteristics of different storage devices to store client files. • External HDDs: ○ use spinning platters and read/write heads which limit how small they can be manufactured ○ can be heavier than other forms of storage media ○ can be damaged if the laptop is knocked or dropped when in use ○ slower than other forms of storage media which could affect processing times ○ allows for large data storage ○ cheaper to purchase than SSD. • SSDs: ○ use interconnected flash memory chips that can be built on the motherboard to make more compact ○ lighter than other forms of storage media ○ uses flash storage with no moving parts making the storage media more robust ○ uses less power than an HDD which means longer battery life (do not require the system to run motors to drive HDD spindles, read/write heads) ○ limited / finite number of writes ○ faster than an HDD to process ○ more expensive than other storage media which can result in SSDs being manufactured with less storage • Removable storage (HDD and SSD): ○ no power is needed because they rely on the power source from the device they are plugged into ○ data security is enhanced because removable storage devices come with built in data encryption technology ○ plug and play features make them easy to use ○ use USB interface which makes them compatible with most devices

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	3–4	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	5–6	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2a	Award one mark for identification and one mark for linked explanation, from: Number of cores (1) because this can affect the number of instructions it can process at any time (1) Multiple applications running (1) which can reduce the speed (1) Low network speed (1) which can result in delays loading applications / files (1) Greater data bus width (1) will give better processor performance (1) Cache memory (1) can improve system performance / reduce the time taken for the processor to access data and instructions (1) Clock speed (1) affects the rate the CPU executes instructions / higher clock speed means faster processing (1) Malware/viruses (1) because it can make the performance of the OS slower (1). Defragmentation (1) puts data/files in a continuous manner to allow the computer to read/retrieve files quicker / increased overall performance of the PC (1) Amount of RAM/memory (1) will affect the amount of available virtual memory / makes the OS quicker/slower (1) Type of storage/SSD (1) loads programs quicker / boot system quicker (1) Accept any other relevant phrasing/wording	2

Question Number	Answer	Mark
2b	Award one mark for each point up to a maximum of four marks. • Detects new peripheral (1) • checks functionality/usability/for errors (1) • Notify user/display message – error or confirmation (1) • Take action (depending on error message)/ enable device (1) • Device drivers (1) • Provides an interface (1) • (Driver) sends/receives signals / communicates with the hardware (1) • (Driver) translates signals so OS and applications can understand (1) • Signals held by buffer / instructions are put in a queue (1) • Instructions are executed (when device is available) (1) ACCEPT: 'requests' for signals DO NOT ACCEPT: Plugging in a device Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
2ci	Award two marks for each identification and two marks for linked explanations, from: Firewall (1) • is software that sits between a device and the internet (1) • monitors incoming / outgoing data packets (1) • blocks data/files that does not comply with set rules (1) Antivirus / anti-malware (1) • scans computer files and email attachments for viruses (1) • looks for suspicious activity / notifies the user (1) • removes or quarantines any viruses that are found (1) VPN (1) • because it encrypts the network connection (1) • sends data packets through a secure tunnel (1) Encryption (1) • so that data is scrambled/encoded (1) • the data is unreadable Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
2cii	Award up to two marks for each of two linked explanations, from: Financial loss (1) • due to legal action / fined / sued / breach of DPA or GDPR legislation • costs to pay experts to fix the system Downtime (1) • loss of education / limited teaching resources available • affects productivity / exams Damage to reputation (1) • Less students would not want to enrol • Difficulty recruiting new staff / might not want to work at the college Accept any other relevant phrasing/wording	4

Question Number	Indicative content
2d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the role of the OS in managing networking, security, memory management and multi-tasking. <u>Networking</u> • Co-ordinates activities of computers, printers, and servers, across a network • Supports protocols that routes data packets around a network • Governs and controls data transmission • Manages and maintains the network using utility programs <u>Security</u> • Authenticates each user on the system by identifying users using the following: ○ Username/password ○ User card/key ○ User attribute such as biometric • Protects the system from threats such as viruses and malware • Controls resources that users are permitted to access <u>Memory management</u> • Allocates memory space to files/programs/clipboard items that are running • Allocates and de-allocates memory before and after execution • Decides how much memory each process will get and when • Monitors each memory location's status to see whether it is free or allocated • Maintain data integrity <u>Multi-tasking</u> • Allows multiple users to perform multiple tasks at the same time, because it allocates each task a portion of the processor time • Keeps track of where the user is in each task • Goes from one task to the other without losing information DO NOT ACCEPT: Reference to having multiple tabs open at the same time, they must say different applications open at the same time, e.g. Word and Excel, Internet and Word.

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	A diagram of a potential system/integration of IT systems and connections used to meet the requirements of the scenario. Example response – note this is indicative only. Credit should be given for a diagram that meets the requirements of the given problem. Diagram may include: Devices: • Servers • Desktop computers • Printer • Data logging system • Tablets • Smart phones Connection types and users: The connection types between devices (wired/wireless) Annotations: The diagram should include annotations for devices and systems as well as the connection types used. The diagram illustrates a network setup. On the left, a box labeled 'Main Office' contains a printer, a switch, a server, and three PCs. The switch is connected to the printer and the three PCs. The server is connected to a 'Router'. The Router is connected to a 'WAP' (Wireless Access Point). Below the Main Office box, there are icons for three smartphones and three tablets. Dashed lines connect the Router to the smartphones and the WAP to the tablets. To the right of the WAP, there are two boxes labeled 'Greenhouses', each containing a 'DATA LOGGING SYSTEM'. Dashed lines connect the WAP to both Greenhouse boxes. At the bottom, a legend shows a dashed line for 'WIRELESS' and a solid line for 'WIRED'.

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Diagram provides partial coverage of devices. • Limited connections are shown • Diagram includes some annotations of the connection types used.
Level 2	3-4	• Diagram provides coverage of mostly appropriate devices. • Connections are shown which meet most requirements of the scenario. • Diagram includes some annotations of connection types used to meet the majority of the requirements of the scenario, but these may not always be the most efficient.
Level 3	5-6	• Diagram provides thorough coverage of appropriate device which fully meet the requirements of the scenario. • Connections are shown which fully meet the requirements of the scenario. • Diagram includes detailed and accurate annotations of devices and connection types used to meet the requirements of the scenario.

Question Number	Answer	Mark
3b	Award one mark for each point up to a maximum of four marks Monitor and control • Data logger / sensors would be used to take readings (1) • At regular intervals (1) • Data sent to data logger (1) • Data is stored in internal memory (of data logger) (1) • Collected data will be time and date-stamped (1) • Transferred to control system (PC) (1) • Readings are compared with required temperature (1) • Temperatures are increased/decreased/changed depending on readings (using smartphones/ tablets/ computers/ automatically) (1) DO NOT ACCEPT: stored to the database (in the question scenario) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
3c	Award one mark for a benefit and one additional mark for an appropriate expansion, for any two of: • Can be set up to monitor over very long periods (24 hours a day) (1) because there are factors that affect humans such as fatigue, boredom (1) • (Readings are) accurate / more precise than human measurements (1) which leads to better informed decision making (1) • Saves time (1) because there is no need to physically visit the greenhouses / set up remotely (1) • Alerts can be sent when there is a change in temperature (1) to enable prompt action to be taken (1) • The system operates in real time (1) which ensures the temperature will always be at its optimum / constant (1) • Systems are robust (1) so they can be used in extreme conditions (1) Additional Guidance Identification and expansion marks may be transposed DO NOT ACCEPT: 'automatically' because it is in the question Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
3d	Award one mark for identification and one additional mark for an appropriate expansion: • Growing /environmental conditions (1) to ensure the conditions in the greenhouses are suitable for the plants / quality of the plants (1) • Historical trends (1) can help predict future / short term decision making (1) • Enables comparison between different greenhouses (1) which could highlight any inconsistencies (1) • To identify anomalies in the data / technology (1) which might need maintenance / replacing the equipment (1) Additional guidance Identification and expansion marks may be transposed. ACCEPT: Examples for decision making Accept any other relevant phrasing/wording	4

Question Number	Indicative content
3e	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners will discuss the benefits and drawbacks to the family of installing open-source software. <u>Benefits</u> • Software is free to use • Can access the source code which allows the family to customise software to suit their needs • They will be able to work with other like-minded users to customise the software • Bugs in the software are spotted and corrected quicker than using proprietary software because the users are not restricted by large organisation's procedures • Can be more reliable because people using the software often spot and correct errors quickly • Ongoing support is free of charge because it is provided by other users/community and not a manufacturer <u>Drawbacks</u> • Compatibility issues can arise between operating systems and applications software and could incur additional costs and time to replace if required to work with each other • Some specialised application software may not run with open source operating systems • May not be as many features included which may make it difficult to carry out specific tasks • Costs might be incurred for additional features • Support relies on the ongoing attention of the users / community which might stop without warning or be sporadic • Must ensure that the software is compatible with the data logging systems • Malware can be introduced by malicious users

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the factors that should be considered when making the sales packs available online. Online access to sales packs: • Compatible with a wide range of platforms (tablets, smart phones, laptops, PCs) • Compatible with operating systems (Windows, iOS) • Consideration of file types and file sizes for downloading Ease of use of sales packs: • Navigation between pages (back, forward, home page) • Drop down menus to select options / filter content Accessibility: • consistency (layout, fonts, colours) • videos with subtitles for hearing impaired/deaf users Cost: • Development costs • Ongoing running costs • Maintenance costs • Staff training costs DO NOT ACCEPT: Responses about zoom, sound options it is the browser that would allow this type of accessibility

Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5–7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8–10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss how the company can ensure the community members find the user interfaces easy to use and feel safe while participating User experience/ ease of use - Navigation between pages - Users will not keep using the site if problems/difficulties are encountered - Must be accessible on a variety of platforms - Ensure the availability of the site (no downtime) - Frequently asked questions Safety: - A moderating team oversees that 'rules of behaviour' are set and adhered to (removed) - Moderating team accepts/rejects/blocks members - Users should be made aware of what is and isn't acceptable to post to avoid any offence or inappropriate material being posted - Privacy of users must be high priority. - Consideration of who is going to be given access (closed group or extended) - Option to remain anonymous - Take measures to ensure the security of users' data when setting up

Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5-8	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	9-12	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate



Llywodraeth Cynulliad Cymru
Welsh Assembly Government

